Requirements Specification

Flash Sale Purchase Platform: Functional and Non-Functional Requirements

1. Scope

The platform lets web and mobile buyers purchase limited-stock electronics (smartphones, laptops, accessories) during timed flash sales, with about 10,000 concurrent buyers. It must prevent overselling, stay responsive under spikes, and keep reservation, payment and order state consistent. Priority: **M** = must have, **S** = should have.

2. Functional Requirements

ID	Area	Requirement	Pri
FR-01	Catalog	Browse products with name, brand, category, price, image; view details and check availability.	M
FR-02	Search	Product search with facets and suggestions (Elasticsearch).	S
FR-03	Pricing	Maintain pricing and offers; price is validated server-side before payment.	M
FR-04	Flash sale	Define a sale per product with start time, end time and total units; expose active status and remaining units; track sold units.	M
FR-05	Cart	Add and remove items, update quantity, calculate subtotal and total.	S
FR-06	Authentication	Authenticate users at the API Gateway before any purchase request is accepted.	M
FR-07	Traffic control	Rate limiting, queue management, token bucket admission and dynamic throttling before requests reach core services.	M
FR-08	Inventory	Track stock per SKU, warehouse and location, with serial number tracking; show availability; reserve and release stock.	M
FR-09	Reservation	On a valid buy request with stock available, create a time-limited reservation (TTL/expiry); support confirm and cancel; expired reservations return stock automatically; track status.	M
FR-10	Idempotency	Reservation and payment accept an idempotency key so retries never create duplicates.	M
FR-11	Payment	Process payment through an external gateway (Stripe/Razorpay); support refund; query payment status.	M
FR-12	Order	Create an order after payment success; track order status; support shipping; recover orders stuck in an intermediate state.	M
FR-13	Purchase response	Return transaction ID, reservation ID and order ID to the client at the end of a successful purchase.	M
FR-14	Failure paths	Return clear results for sold out, rate-limited, payment failed and reservation expired; release the reservation on payment failure.	M
FR-15	Notifications	Send SMS, email or push messages for reservation, payment and order events.	S
FR-16	Logistics	Hand shipped orders to a logistics provider (Delhivery/Blue Dart) and record status.	S
FR-17	Events	Publish domain events to Kafka for async processing, retries and dead-letter handling.	M
FR-18	User actions	Users can place, view and cancel orders.	M

3. Non-Functional Requirements

ID	Category	Requirement	Proposed measure
NFR-01	Scalability	Support about 10,000 concurrent buyers; stateless services scale horizontally; load balancer auto-scales at L4/L7.	Load test at 10,000 users and above
NFR-02	Performance	Fast reservation path under load; hot reads and stock counters served from Redis; search from Elasticsearch.	p95 buy request under 500 ms (to be agreed)
NFR-03	Consistency	No overselling; atomic stock decrement; reservation TTL; idempotent payment and order creation.	Sold units never exceed total units
NFR-04	Availability	Health checks, redundant instances, graceful degradation through queueing and throttling.	99.9% during sale window (to be agreed)
NFR-05	Reliability	Kafka retries and dead-letter queue; order recovery; compensation when a step fails.	No paid order left without an order or refund
NFR-06	Security	HTTPS on all links; DDoS protection and bot filtering at the edge; authentication at the gateway; verified payment callbacks.	No unauthenticated purchase
NFR-07	Data integrity	PostgreSQL is the source of truth for product and inventory; Redis counters can be rebuilt from it.	Reconciliation shows zero drift

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NFR-08	Maintainability	Independent microservices with clear boundaries, communicating over REST and events.	Deploy a service without others
NFR-09	Observability	Health checks, status tracking, correlation ID across services, event audit trail, alerts on failures.	Every purchase traceable end to end
NFR-10	Fault tolerance	Timeouts, retries with backoff, circuit breaker on external calls, especially the payment gateway.	Gateway outage does not stop browsing

4. Constraints and Assumptions

Technology: PostgreSQL (Product DB, Inventory DB), Redis (cache, rate limits, sessions), Elasticsearch (search), Kafka (events). **Integration:** Stripe or Razorpay, SMS/email/push provider, Delhivery or Blue Dart. **Assumptions:** measures marked "to be agreed" are proposals for the team to confirm; the cart is optional for the core flash-sale path.